Marco Del Fava

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Research Profile

Biomedical Engineering M.S. student at UC Davis and research volunteer in Dr. Jie Zheng's lab, working on human single-unit electrophysiology. Builds research software for neural acquisition and spike analysis, with interests in Intan acquisition, real-time spike sorting, and semantic selectivity in single-neuron responses.

Research Experience

Research Volunteer, UC Davis Health — Z-Lab (Dr. Jie Zheng) — Jul 2025 - Present

- Develop research workflows connecting acquisition, spike processing, and downstream single-neuron response analysis.
- Analyze human epilepsy microwire recordings using O-Sort and SpikeInterface, with attention to waveform structure and unit curation.

Selected Projects

Intan acquisition and experiment synchronization

Connect neural recordings to reproducible processing and experiment timing.

- Implemented Intan RHD loading, probe configuration, and SpikeInterface/Kilosort4 processing in zlab_sorting_script.
- Developing Intan RHX/DeepArray acquisition and event-synchronization integration; hardware bench validation remains pending.

Status: File-processing workflow implemented; live acquisition and timing integration in progress.

Intan workflow

Waveform discrimination and online sorting research

Preserve unit-specific waveform information while investigating streaming analysis.

- Implemented WeightBank waveform representations in spike_discrim; evaluated post-hoc clustering against putative O-Sort unit labels.
- Developed synthetic streaming/detection infrastructure in realtime-sorter-mvp and experimental competitive-learning sorting in SNNeuro.

Status: Discrimination evaluated offline; online unit sorting remains experimental, without validated end-to-end latency.

Discrimination results · Streaming, not unit sorting · Experimental sorter

Single-neuron response analysis and semantic tuning

Investigate relationships between stimulus meaning and single-neuron responses.

- Implemented a seeded NEURON prototype mapping sentence similarity to stochastic input rates, with fixed synaptic weights.
- Developed semantic-trajectory tooling in ZEmbeddings; exploring stimulus-aligned response analysis and adaptive stimulus selection.

Status: Simulation and text analysis implemented; adaptive experiments proposed, not demonstrated human concept selectivity or causality.

Simulation and limitations · Semantic text analysis

Selected Projects (continued)

Supporting acquisition work - Neuralynx / Dark Horse Neuro

Configure acquisition sessions and replay archived Neuralynx recordings.

- Implemented DHN-AQ RC/CS configuration writers and SSH launch orchestration.
- Developed an NRD-to-UDP replay prototype for DHN-AQ ingestion.

Status: Prototype implemented; hardware reception and recording validation remain in progress.

Acquisition implementation

Education

M.S. Biomedical Engineering, University of California, Davis — Fall 2026 - Present

Python for Data Science, Professional Concentration, UC Davis Continuing and Professional Education — 2025

M.S. Psychology, Arizona State University — 2023

B.S. Neuroscience, Physiology, and Behavior, University of California, Davis — 2017

Technical Skills

Neural Data & Signal Processing

- Waveform features, spike-vs-noise evaluation, and afferent clustering
- SpikeInterface preprocessing, Kilosort4, and manual unit curation workflows

Neural Acquisition & Tooling

- Intan RHD loading and probe-aware sorting workflows
- DHN-AQ RC/CS configuration, SSH orchestration, and NRD replay prototypes

Machine Learning & Data Science

- Python (NumPy, Pandas, scikit-learn, PyTorch, Numba); MATLAB
- Sentence embeddings and seeded NEURON simulation prototypes

General Technical

- Git, Linux, SSH, shell scripting, Docker, and AWS GPU workflows
- Microsoft 365 administration; SQL and PostgreSQL

Additional Professional Experience

IT Administrator, First Step Communities — Jul 2021 - Present

- Managed Microsoft 365 IT infrastructure across 4 shelter sites, including user provisioning, device management, and systems coordination.

Analyst, Eurofins Test America — Oct 2019 - Jun 2021

- Maintained high-accuracy data entry and review in a regulated laboratory environment; supported QA documentation and adherence to standard operating procedures.

Certifications

- Data Analysis with Matplotlib, Pandas, and Scikit-Learn Python Libraries — UC Davis Continuing and Professional Education (2025-09-18)
- Biomedical Research; Research Administration; GCP Social and Behavioral Research — CITI Program